

around an NE555 timer, a decade counter CD4017, AND, NOR and NOT gates, and a few discrete components.

NE555 is used in astable mode to generate the basic clock for decade counter CD4017. The clock frequency of NE555 is determined by resistor R1,

preset VR1 and capacitor C1. It can be adjusted by varying preset VR1. Pin 5 of timer NE555 is the control voltage pin that is primarily used for filtering when the timer is used in noisy environments. However, by applying a voltage at this pin, it is possible to vary the calculated period.

A 0.01 μ F capacitor connected to pin 5 of NE555 bypasses any noise from altering the calculated pulse width. Reset

pin 4 of NE555 is pulled high through resistor R3. When switch S1 is closed, pin 4 is connected to ground and the astable multivibrator stops clock generation.

Decade counter CD4017 is at the heart of the traffic light system. It is a 5-stage Johnson counter having ten decoded outputs. The clock input receives the clock pulse from pin 3 of NE555. Schmitt trigger action in the clock input circuit of CD4017 provides pulse shaping that allows unlimited clock input pulse rise and fall times. These counters advance by one count at the positive clock signal transition if clock-inhibit signal pin 13 is low. A high signal on reset pin 15 clears the counter to zero.

Q0 through Q7 outputs of CD4017 control the traffic light signal for four roads as shown in Fig. 2. Q0 and Q1 outputs control the green, orange and red traffic light sequence for a single road (say, road 1).

When reset switch S2 is pressed, Q0 output of CD4017 is high and the other outputs are low. The low Q1 output is inverted by NOT gate N13 and fed to pin 13 of AND gate N5. Q0 output is fed directly to pin 12 of gate N5. Both the high inputs of AND gate N5 make its output high and the green LED (LED1) glows.

Pin 14 of CD4017 receives a clock pulse and its Q1 output goes high while the other outputs are low. Q1 output is fed directly to pin 8 of gate N6, while the low Q0 output is inverted and fed to pin 9 of gate N6. Both the high inputs of AND gate N6 make its output high and the orange LED (LED3) glows.

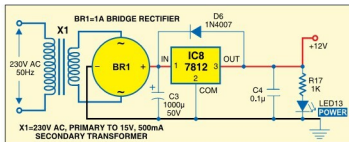


Fig. 3: Power supply circuit

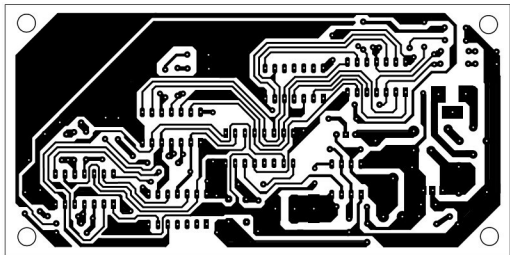


Fig. 4: Actual-size, single-side PCB for the adaptive traffic light system

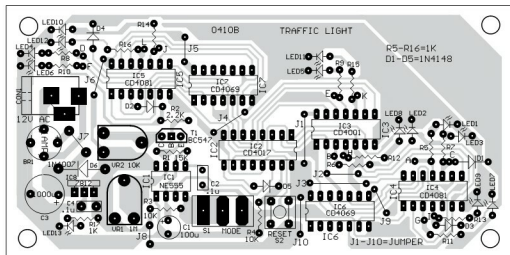


Fig. 5: Component layout for the PCB